Descriptive Geometric Morphometrics in a Browser: Statistical Architecture, Synthetic Demonstration, and Reproducible Interpretation

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A synthetic, descriptive computing demonstration

Abstract

This technical report documents a serverless demonstration of descriptive geometric morphometrics implemented with NumPy and SciPy inside a Pyodide WebAssembly runtime. The application accepts either one of three deterministic synthetic configurations, a de-identified matrix, or an explicitly detected local photo mesh sampled into 68 ordered two-dimensional landmarks. It removes translation and centroid size, aligns configurations by proper singular-value-decomposition rotations, constructs a 132-dimensional Kendall tangent representation, calculates principal-component scores, and reports Procrustes, shrinkage-regularized Mahalanobis, and residual distances.

The implementation is intentionally descriptive. Reference A, Reference B, and their pooled ensemble are simulations, not estimates of biological populations. Optional photo landmarking and texture warping occur only after an explicit action and remain in browser memory; Python receives coordinates, not pixels. A proportional warp visualizes the same residual field used by the numerical output. The worked pooled-reference Blend example yields partial Procrustes distance 0.06441, full Procrustes distance 0.06437, regularized Mahalanobis distance 9.759, residual root mean square 0.00781, and Geometric Displacement Index 0.46 of 10. These are uncalibrated geometric magnitudes rather than percentiles, probabilities, classifications, or appearance ratings.

Keywords

geometric morphometrics, Generalized Procrustes Analysis, tangent space, principal component analysis, covariance shrinkage, Mahalanobis distance, WebAssembly, local face landmarks, piecewise-affine warp, reproducible simulation

Descriptive Geometric Morphometrics in a Browser

Geometric morphometrics studies configurations of corresponding landmarks while preserving the geometry among those landmarks. Unlike a list of isolated lengths, an ordered coordinate configuration retains information about relative position, joint displacement, and multivariate shape. The central statistical difficulty is that raw coordinates also contain arbitrary translation, overall scale, and rotation. Those nuisance transformations must be separated from the shape signal before distances or covariance summaries are meaningful (Dryden & Mardia, 2016; Rohlf & Slice, 1990).

The present application turns this sequence into an inspectable browser workflow. A configuration is validated, centered, scaled to unit centroid size, rotated into a reference frame, projected into a tangent plane, and compared with a structured simulated ensemble. Each transformation is deterministic. JavaScript manages the interface and transfers only numeric coordinates to Python; the Python module performs the linear algebra and returns strict JSON.

This report has two purposes. First, it states exactly what every displayed number means and how it is calculated. Second, it establishes the inferential boundary. The software does not operationalize attractiveness, health, identity, sex, gender, ancestry, or clinical status. It does not optimize a configuration toward a mean. Its output is a reproducible demonstration of shape-space mechanics, consistent with the distinction between geometric description and substantive biological inference emphasized by Adams et al. (2004) and Zelditch et al. (2012).

Scientific Scope, Unit of Analysis, and Estimand

The unit of analysis is one 68 by 2 matrix X whose rows are ordered landmark coordinates. A valid row has one horizontal and one vertical coordinate, and every configuration must use the same row-to-feature correspondence. The engine's estimand is descriptive separation between the analyzed configuration and a selected simulated reference consensus after translation, uniform scale, and proper rotation have been removed.

Three families of output address different geometric questions. Procrustes distances summarize unweighted separation in aligned unit shape space. Tangent coordinates and PCA scores describe direction relative to the simulated reference covariance axes. Regularized Mahalanobis distance weights the tangent displacement by the inverse simulated covariance, so a displacement along a low-variance direction contributes more than an equally sized displacement along a high-variance direction (Mardia et al., 1979). Residual vectors retain landmark-level direction, and residual RMS compresses their magnitudes into one scalar.

None of these quantities has an empirical sampling interpretation here. The reference covariance is generated rather than observed, the shrinkage amount is estimated within that simulation rather than validated against an external objective, and the analyzed configuration is not sampled from a declared population. Consequently, a distance cannot be labeled typical, unusual, desirable, or undesirable. The correct conclusion is comparative and conditional: for this input, this simulated reference, and this implementation, the reported magnitude equals the stated calculation.

Serverless and Local-First Statistical Architecture

The application is a static GitHub Pages site. HTML defines the accessible three-tab workspace, CSS provides the responsive phone-like presentation, JavaScript controls state and validation, and Pyodide instantiates Python inside the browser. NumPy and SciPy are loaded from a pinned content-delivery-network path. No application server, database, session store, or telemetry endpoint is required.

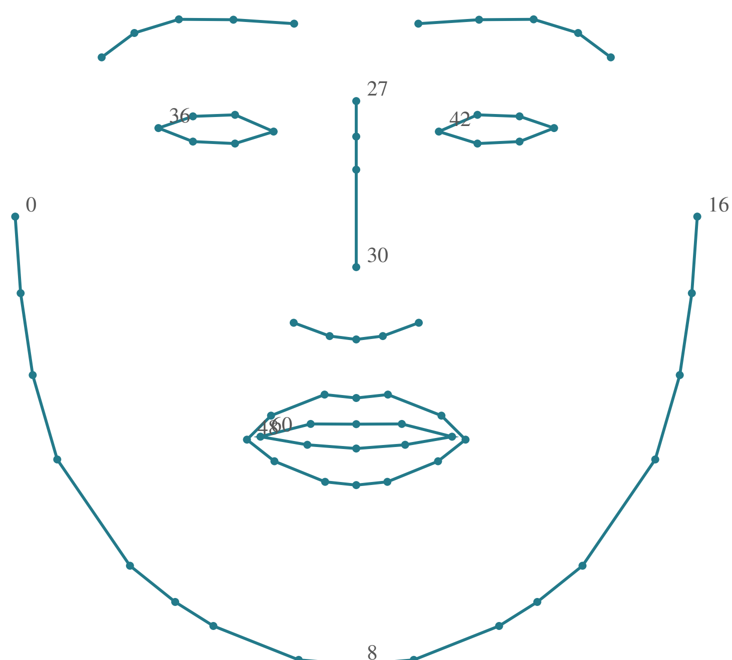
1	2	3	4	5
Acquire	**Normalize**	**Align**	**Project**	**Describe**
local 68 x 2 values	center and unit scale	proper SVD rotations	132 tangent values	distances and residuals

A selected JPG, PNG, or WebP file receives a temporary browser object URL for viewing, panning, zooming, and rotation. That object URL is revoked when the image is replaced, removed, or the page closes. No detection occurs until an explicit button press. The pinned MediaPipe task then obtains one dense mesh in browser memory and a fixed adapter samples 68 vertices. Blendshape outputs are disabled, and pixels are never sent to Python or an application server.

For an analysis, JavaScript creates a Float64Array of 136 values and assigns it to the Pyodide global namespace. The top-level Python function validates and reshapes the values, fits or retrieves the selected cached reference model, and returns JSON. A photo-derived run uses only those 136 numbers in Python. Returned residual vectors may drive a browser-canvas texture warp, while exports omit pixel data. Synthetic and de-identified coordinate routes remain independent alternatives.

The 68-Point Coordinate Schema

Landmark methods require correspondence: row j must represent the same labeled location in every configuration. The application follows the conventional 68-point ordering used by the Dlib schema: 0-16 jawline, 17-21 right brow, 22-26 left brow, 27-30 nose bridge, 31-35 nose base, 36-41 right eye, 42-47 left eye, 48-59 outer lip, and 60-67 inner lip. These labels establish topology only; they do not make the synthetic template a biometric standard.



Open feature paths such as the jaw and brows are drawn without connecting their endpoints. Eye and lip paths are closed. Coordinate files may be JSON, CSV, or plain text, but after parsing they must contain exactly 136 finite numbers. The photo route uses a fixed list of 68 unique MediaPipe mesh indices as an engineering correspondence; it is not native Dlib detector output. Template A samples those same indices from MediaPipe's canonical OBJ at commit a908d668, retains x and y, reverses the image y-axis, then centers and unit-scales the result. The Apache-2.0 provenance is recorded in THIRD_PARTY_NOTICES.md. This source match does not make the adapter a validated measurement model. Missing or reordered landmarks invalidate direct comparison (Bookstein, 1991; Zelditch et al., 2012).

Translation Removal and Centroid Size

Let X contain k = 68 landmark rows. The centroid is the row-wise average x-bar, and the centered configuration is Xc = X - 1(x-bar)^T. Subtracting the same centroid from every row removes the two degrees of freedom associated with horizontal and vertical translation. The operation preserves all pairwise landmark differences.

$$c(X) = ||Xc||_F = \sqrt{ \sum_j ||x_j - \bar{x}||_2^2 }$$

Centroid size c(X) is the Frobenius norm of the centered configuration. It is not face area, image resolution, or a biological size estimate. It depends on the units of the submitted coordinates and on the complete landmark configuration. The reported centroid-size card records this value before normalization so the user can audit the scale that was removed.

$$Z = Xc / c(X), \quad ||Z||_F = 1$$

Dividing by centroid size produces a unit preshape Z and removes one uniform-scale degree of freedom. A configuration with zero or numerically negligible centroid size is rejected because normalization would be undefined. Translation and scale invariance are tested by applying arbitrary shifts and positive multipliers to an input and confirming that its post-alignment shape distances remain equal within floating-point tolerance (Dryden & Mardia, 2016).

Generalized Procrustes Analysis

Generalized Procrustes Analysis aligns an ensemble of preshapes to a jointly estimated consensus. If Zi is configuration i, Ri is a proper two-dimensional rotation, and M is a unit-norm consensus, the engine minimizes the summed squared Frobenius residuals over all rotations and the consensus.

$$\min_{(M,R_i)} \sum_i \|Z_i R_i - M\|_F^2, \quad \det(R_i) = 1$$

The algorithm begins with the first preshape as the working frame. Every preshape is rotated toward that frame, the aligned configurations are averaged, the mean is recentered and normalized, and the new mean is aligned back to the previous frame. These steps repeat until the consensus change is below 1e-10 or 100 iterations have been attempted. A final alignment and normalized average ensure that the returned ensemble and consensus share one frame.

This is a multi-configuration fit rather than a single pairwise alignment. The reported generalized sum of squares is the sum of squared configuration-to-consensus distances. The application also reports the mean and maximum reference distances, the number of iterations, and whether the convergence tolerance was met. Gower (1975) established the generalized formulation, and Rohlf and Slice (1990) developed the iterative landmark superimposition approach used throughout modern geometric morphometrics.

Proper Rotation by Singular Value Decomposition

For one preshape Z and a current reference M, the rotation problem is solved from the 2 by 2 cross-product $H = Z^T M$. SciPy computes the singular value decomposition $H = U S V^T$. The orthogonal matrix $U V^T$ minimizes the least-squares discrepancy, subject to a determinant correction.

$$R = U \, \mathrm{diag}[1, \det(U V^T)] \, V^T$$

If $\det(U V^T)$ is negative, the unconstrained solution includes a reflection. The engine changes the final singular-vector sign so $\det(R) = 1$. This restriction matters because a reflected configuration reverses handedness and should not be treated as an ordinary planar rotation. The aligned preshape is $Y = ZR$.

SVD is numerically preferable to solving for an angle through ad hoc trigonometric expressions. It supplies the least-squares orthogonal factor directly and generalizes to higher-dimensional Procrustes problems. The implementation uses scipy.linalg.svd with finite-value checking handled during input validation. Proper rotation invariance is tested by multiplying an input by a known rotation matrix and confirming that the resulting distance measures agree with those from the original configuration (Goodall, 1991).

Consensus Convergence and Alignment Diagnostics

A small consensus change is a numerical stopping condition, not proof that a reference represents an external population. At iteration t the engine evaluates the Frobenius norm $\|M(t+1) - M(t)\|$. Convergence is recorded when that norm is less than 1e-10. If the maximum of 100 iterations is reached first, the analysis still returns a status that exposes the failure.

$$G^* = sum\_i \, ||Yi - M\text{-}hat||\_F\hat{}2$$

The diagnostics shown in the interface serve distinct purposes. Iterations describe algorithmic effort. Reference n gives 160 for Reference A or B and 320 for the pooled reference. The generalized sum of squares measures total within-reference aligned dispersion. The covariance condition number, calculated later, describes sensitivity of linear solves after shrinkage and ridge stabilization.

These diagnostics should be read before interpreting distances. A nonconverged GPA, nonfinite output, wrong tangent dimension, or failed Cholesky factorization would make downstream summaries unreliable. In the deterministic pooled model used for the worked example, GPA converges in four iterations. Fixed RandomState seeds stabilize the simulated draw stream, but SVD and eigendecomposition remain numerical-library dependent. The regression fixture therefore uses a tight tolerance, and exact artifact reproduction requires the pinned development stack.

Preshape Sphere and Kendall Shape Space

After centering and unit scaling, a planar configuration lies on a preshape sphere. However, two preshapes that differ only by a proper planar rotation represent the same shape. Kendall shape space is therefore a quotient space: rotational orbits on the preshape sphere are identified. This curvature is why ordinary multivariate analysis cannot be applied naively to raw coordinates (Kendall, 1984).

For k planar landmarks, the unconstrained coordinate vector has 2k entries. Removing two translations, one uniform scale, and one planar rotation leaves 2k - 4 shape degrees of freedom. With k = 68, the resulting dimension is 132. The interface reports this number as a structural check.

dimension = 2(68) - 2 translation - 1 scale - 1 rotation = 132

Statistical calculations proceed in a tangent plane at the fitted consensus rather than directly on the curved quotient. This is a local approximation. It is reliable when configurations remain sufficiently close to the tangent point, but it should not be treated as a globally distance-preserving map. The engine guards against a nearly orthogonal preshape and consensus because central projection would then divide by a value near zero (Dryden & Mardia, 2016; Klingenberg & Monteiro, 2005).

Constructing the 132-Dimensional Tangent Basis

The engine explicitly constructs four nuisance directions in the 136-dimensional ambient coordinate vector. Translation-x alternates 1 and 0, translation-y alternates 0 and 1, the radial direction equals the vectorized consensus, and the infinitesimal rotation direction rotates every consensus row by 90 degrees.

N = [t_x, t_y, vec(M), vec(JM)],   B = canonical MGS(N-perp)

The engine first orthonormalizes the four nuisance directions, then applies modified Gram-Schmidt with one reorthogonalization pass to the 136 standard basis vectors in index order. The accepted 132 columns form B, with B^T B = I and B orthogonal to each nuisance direction. This fixes the chart orientation conditional on the fitted consensus instead of accepting an arbitrary null-space basis.

For aligned preshape Y, let a be the inner product between vec(Y) and vec(M). Central projection first forms vec(Y)/a - vec(M), which lies in the affine tangent plane at M. Multiplication by B^T gives the nonredundant coordinate vector z. Because B is orthonormal, Euclidean operations on z correspond to operations within the selected tangent subspace. The basis depends on the selected reference consensus, so changing Reference A, Reference B, or Pooled A+B recomputes the coordinate system.

z = B^T [vec(Y)/<vec(Y),vec(M)> - vec(M)]

Tangent-Space Principal Component Analysis

PCA summarizes covariance directions in the simulated tangent sample. Let zi denote reference configuration i and z-bar its sample mean. The sample covariance is the sum of outer products of centered tangent vectors divided by n - 1. It is symmetric and positive semidefinite.

S = (1/(n-1)) sum_i (zi - z-bar)(zi - z-bar)^T = V Lambda V^T

The engine uses scipy.linalg.eigh, which is designed for symmetric matrices. Eigenpairs are sorted from largest to smallest eigenvalue, and small negative values caused by floating-point roundoff are truncated to zero. For an analyzed tangent vector z, score j is vj^T(z - z-bar). The displayed percentage is lambda-j divided by the sum of all 132 eigenvalues (Jolliffe, 2002).

A PCA sign is arbitrary: multiplying an eigenvector and its score by -1 describes the same axis. Therefore, a positive score is not better than a negative score. Score magnitude describes coordinate displacement along that simulated axis, while the percentage describes how much simulated reference variance the axis accounts for. For stable presentation, each eigenvector is signed so its largest-magnitude loading is positive; exactly repeated eigenvalues can still leave a multidimensional eigenspace without a unique axis orientation. The JSON response returns the first 10 scores, eigenvalues, and ratios.

Structured Synthetic Reference Ensembles

A diagonal noise model would make landmarks vary independently and would fail to demonstrate covariance-aware geometry. Instead, the engine defines 12 smooth tangent deformation modes spanning jaw width and depth, brow position and curvature, eye width and height, nose width and length, lip width and height, bilateral asymmetry, and a local lower-contour mode. Each raw mode is projected into the 132-dimensional tangent subspace and normalized.

For every simulated configuration, independent normal coefficients multiply the 12 modes at fixed amplitudes. Small landmark-level normal noise is also projected into the tangent subspace. The perturbed configuration is returned to unit preshape form, then random translation, positive scale, and rotation are added as nuisance changes. GPA must recover from these intentionally introduced transformations.

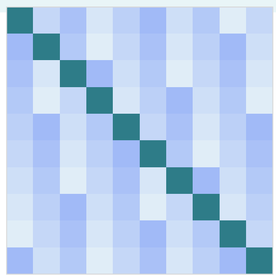
Reference A uses 160 shapes around Synthetic Template A with seed 31041. Reference B uses 160 around Synthetic Template B with seed 72107. The pooled model concatenates both ensembles and therefore has n = 320. This construction produces cross-landmark covariance and a stable legacy random stream within the declared stack. It is not fitted to Dlib users, a biological group, or any empirical sample. Synthetic structure supports software demonstration, not population inference (Bishop, 2006).

Covariance Estimation, Shrinkage, and Ridge Stabilization

The tangent sample covariance S contains 132 by 132 entries. Even with 160 or 320 simulated shapes, correlations can make S ill-conditioned. Direct inversion would amplify numerical error and could make Mahalanobis distance unstable. The application therefore shrinks S toward its own diagonal rather than toward a scalar identity matrix.

Sigma-hat = (1 - lam) S + lam diag(S) + epsilon I

lam = clip_[0,1]{ sum_(i != j) Var-hat(s_ij) / sum_(i != j) s_ij^2 }*



Schematic covariance structure: diagonal variance plus retained cross-coordinate dependence

Diagonal-target shrinkage preserves coordinate-specific variances while reducing off-diagonal magnitude by lam. The intensity is not fixed by hand: lam is the analytic minimum-risk estimate for this target, which for the pooled reference is 0.0320. The ridge epsilon equals 1e-6 times the average post-shrinkage variance, bounded below by machine precision, and the matrix is symmetrized before Cholesky factorization (Ledoit & Wolf, 2004; Schafer & Strimmer, 2005).

The diagnostics report the ridge, the condition number, and the fraction of the Frobenius norm carried by off-diagonal entries. Those quantities show that covariance structure remains present and that the distance is not merely rescaled Euclidean distance.

Regularized Mahalanobis Distance

Let z be the analyzed tangent coordinate and z-bar the simulated tangent mean. Their difference delta = z - z-bar is measured relative to the regularized covariance. The squared Mahalanobis distance is a quadratic form.

D_M^2 = delta^T Sigma-hat^(-1) delta,   D_M = sqrt(max(0,D_M^2))

The engine does not explicitly form Sigma-hat inverse. SciPy first computes a lower Cholesky factor and then solves the two triangular systems needed for Sigma-hat inverse times delta. The final inner product delta^T solution yields the squared distance. A maximum with zero protects the square root from a tiny negative value caused by roundoff.

Mahalanobis distance changes the geometry: displacement along a direction with small simulated variance receives greater weight, while displacement along a direction with larger simulated variance receives less. However, the value is not compared with a chi-square distribution because the reference is simulated, regularized, and not linked to a sampling model for the input. The displayed value therefore answers only how large this tangent displacement is under the selected simulated covariance metric (Mahalanobis, 1936; Mardia et al., 1979). The returned reference position is an in-sample rank among simulated reference distances, not a population percentile or p-value; held-out simulations serve only as a numerical scale check.

Partial and Full Procrustes Distances

After optimal rotation, the partial Procrustes distance is the Frobenius norm between aligned input Y and consensus M. It is also called a chord distance because it measures the straight chord between unit preshapes in the embedding space.

$$d_P(Y,M) = ||Y - M||_F$$

The full Procrustes measure allows an additional scalar adjustment. With unit preshapes and inner product $a = \langle vec(Y), vec(M) \rangle$, the implemented expression is the square root of $\max(0, 1 - a^2)$. Both measures are nonnegative and approach zero as the aligned configurations agree.

$$d_F(Y,M) = \sqrt{\max(0, 1 - a^2)}$$

These measures should be compared only under the same landmark schema and preprocessing definition. They contain no covariance weighting and no automatic threshold. A smaller value means closer geometric agreement with the selected consensus in its defined shape space; it does not mean a more desirable, healthy, or representative appearance. The proximity of partial and full values in the worked examples reflects the small local separations after unit normalization (Dryden & Mardia, 2016; Klingenberg & Monteiro, 2005).

Landmark Residuals and Root Mean Square

The aligned residual at landmark j is $r_j = M_j - Y_j$. Each r_j is a two-dimensional vector pointing from the analyzed aligned coordinate toward the selected simulated consensus coordinate. Its Euclidean length records landmark-level separation in unit shape space. Multiplying the aligned residual matrix by the transpose of the fitted rotation also expresses the same vectors in normalized input orientation.

$$r_{RMS} = \sqrt{(1/68) \sum_j \|r_j\|_2^2}$$

Residual RMS aggregates the 68 vector magnitudes. It is sensitive to distributed small differences and to a few larger differences, but it discards direction and regional identity. The canvas preserves direction by drawing arrows, while the JSON retains both vector components and all 68 magnitudes.

The arrows can be rendered at 1x, 3x, or 6x. For photo landmarks, multiplying the input-orientation residual by centroid size restores pixel units before rendering. This occurs in JavaScript after Python returns; RMS and every other statistic remain unchanged. Residuals diagnose where aligned coordinate systems differ. They are not suggested edits, and the engine contains no optimization routine.

Sample A, Sample B, and the Synthetic Blend

The three buttons in Research Input select teaching configurations rather than categories of people. Sample A is the first shape generated around Synthetic Template A from seed 90011. Sample B is the first shape generated around the deliberately altered Synthetic Template B from seed 90021. Each includes structured tangent variation plus nuisance translation, scale, and rotation.

base_blend = preshape(0.55 Template_A + 0.45 Template_B)

Blend begins with a landmark-by-landmark convex combination of the two synthetic templates. The 55-to-45 weighting is explicit in core/analytics.py. The combined coordinates are centered and scaled to a unit preshape, then a new deterministic configuration is simulated around that base with seed 90031. The displayed Blend is therefore not the exact midpoint of the displayed Samples A and B.

The input choice and reference choice answer different questions. Selecting Blend chooses the configuration being analyzed. Selecting Pooled A+B chooses a reference ensemble containing 160 A-based and 160 B-based simulated shapes. Selecting Reference A or B changes the consensus, tangent basis, covariance, PCA axes, and distances. None of these three sample buttons blends a photograph; the explicit local-photo route is a separate input control and does not produce an appearance score.

Worked Deterministic Example

Table 1 records engine version 2.0.0 results for all three built-in configurations against Pooled A+B. The values were produced by run_pipeline_from_js with the fixed seeds documented above. Rebuilding the report imports the current analytics module and verifies the Blend values before writing the PDF.

Input	Partial	Full	Mahalanobis	Centroid	RMS
Sample A	0.05457	0.05455	10.201	0.896	0.00662
Sample B	0.05000	0.04998	12.765	0.898	0.00606
Blend	0.06441	0.06437	9.759	0.840	0.00781

Note. All reference configurations are simulated.

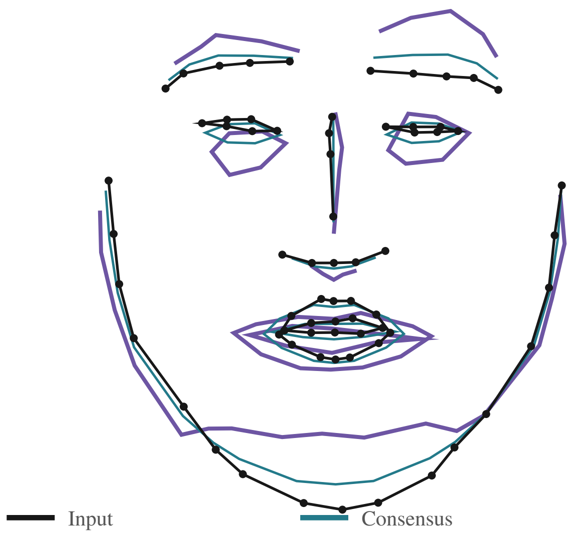
For Blend, the pooled reference GPA converges in 4 iterations with n = 320. The partial distance 0.06441 and RMS 0.00781 summarize aligned unit-space separation. Regularized Mahalanobis 9.759 is larger numerically because it measures the same tangent displacement in standardized covariance geometry; it is not on the same scale as a Procrustes distance. PC1 accounts for 27.3% of pooled simulated variance and the Blend PC1 score is -0.00741. Rescaling its partial distance by 10 min(1, dP / sqrt(2)) gives a Geometric Displacement Index of 0.46.

The only defensible conclusion is that the generated Blend differs from the pooled simulated consensus by the reported amounts under this deterministic pipeline. The table cannot establish typicality, preference, identity, or a biological group difference. It is a reproducibility fixture that helps readers trace interface values back to equations and code.

Landmark and Photo Texture Warp Visualization

The proportional warp makes the residual field visible. In aligned shape space, Wj(s) = Yj + s(Mj - Yj). For a photo input, the residual is rotated back to input orientation and multiplied by centroid size, giving the pixel destination xj' = xj + s c(X) rj. The selected scale changes rendering only.

GDI = 10 min[1, d_P / sqrt(2)]



The shape-space contour and the photo texture are rendering aids, not new statistical models. The photo renderer adds eight fixed boundary anchors, computes a Delaunay triangulation, and uses one affine pixel map per triangle. Vertex shifts are capped at 16% of the face-box diagonal, and a line search reduces the requested scale if a triangle would reverse orientation.

Original, Split, and Warped views use the same locally decoded image. Pixels are not uploaded or placed in the JSON export. Changing warp scale or mesh visibility redraws the canvas without changing any metric. The render is a geometric comparison, not a recommendation, prediction, improvement, or appearance evaluation.

Study-Context Annotations and Future Research

Environmental Stress Index, Pathogen Prevalence Index, and Operational Sex Ratio are optional metadata sliders. The first two range from 0 to 1, and the third ranges from 0.50 to 1.50 around a neutral code of 1.00. Their values are stored under study_context_metadata with analytic_role equal to annotation_only.

Moving a slider never changes GPA, the tangent basis, PCA, covariance, distances, the Geometric Displacement Index, or residuals. The interface says this explicitly because adding a context variable to a screen can otherwise create the false impression that a causal or associational model has been fitted. The current values have no operational definition outside a future protocol. In particular, an operational ratio would require a declared numerator, denominator, population, and observation window.

A genuine bio-social study would need a target population, sampling frame, measurement protocol, rater reliability analysis, missingness plan, predeclared estimand, and an outcome that is scientifically and ethically defensible. Repeated observations would require within-subject dependence to be modeled. Multivariate distance-based regression could relate a distance matrix to covariates, but permutation structure and dependence must be respected (Anderson, 2001; McArdle & Anderson, 2001; Shehzad et al., 2014). Those requirements are beyond this synthetic demonstration.

Numerical Safeguards and Runtime Diagnostics

Input validation requires exactly 136 finite values, a 68 by 2 shape after reshaping, and positive centroid size. Photo files and coordinate files have separate accepted extensions and size limits. Malformed JSON, nonnumeric tokens, NaN, infinity, near-zero geometry, centroid-size overflow, invalid reference keys, and inappropriate tangent projection are returned as bounded error messages.

- GPA prohibits reflections and exposes convergence status and iteration count.
- The canonical tangent basis must have exactly 132 orthonormal columns.
- Covariance is shrunk, ridge-stabilized, symmetrized, and Cholesky-factorized.
- Linear solves replace an explicitly constructed covariance inverse.
- Reference-range and tangent-chart warnings accompany extrapolating inputs.
- The reported reference position is explicitly labeled in-sample and simulated.
- JSON rejects nonfinite values, and fixed RandomState seeds stabilize test streams.
- Photo shifts are capped and triangle orientation is protected by a scale line search.

The covariance condition number is a sensitivity diagnostic rather than a pass-fail proof. A very large value would suggest that small numeric perturbations could change the solution. The off-diagonal Frobenius fraction confirms that the covariance retains structured dependence. These values should be interpreted jointly with reference n, ridge magnitude, and GPA convergence.

At the web layer, the runtime modal reports WebAssembly initialization, NumPy/SciPy loading, and analytics-module loading separately. Controls are disabled while an analysis is running to prevent state races. Photo landmarking is an explicit opt-in action; a fixed 68-index map is validated before use. Temporary Pyodide globals are deleted in a finally block, and an analysis token prevents an obsolete asynchronous response from overwriting a newer configuration.

Validation, Invariance, and Reproducibility

A reproducible implementation must test properties, not only example outputs. Translation invariance is checked by adding a constant vector to all landmarks; uniform-scale invariance by multiplying every centered coordinate by a positive constant; and rotation invariance by applying a proper 2 by 2 rotation. Partial and full Procrustes distances, tangent summaries, and covariance-aware distance should remain equal within floating-point tolerance.

Schema tests verify 68 residual vectors, 132 tangent coordinates, 10 returned PCA summaries, finite JSON, and explicit simulated-reference metadata. Negative tests cover wrong row counts, all-equal coordinates, nonfinite entries, reflection, invalid reference names, and malformed browser files. Interface tests verify direct panning from image pixels, fit/reset behavior, opt-in landmark status, 68 unique mapped mesh indices, triangulation, Original/Split/Warped redraws, the exact index formula, and the absence of image pixels from the Python and export payloads.

Reproducibility also depends on provenance. The engine version, schema version, reference key, reference n, and deterministic seeds are returned with each result. Seventy-seven tests cover invariances, algebraic identities, the canonical template digest, tangent-basis properties, malformed inputs, warning paths, and the browser JSON contract. The report builder verifies its worked fixture and exact 27-page count before printing a SHA-256 digest. requirements-dev.txt pins the authoring stack; neither seeds nor sign conventions alone promise cross-platform bitwise equality.

Discussion, Limitations, and Conclusion

The application demonstrates that sophisticated shape statistics can run entirely in a static browser site while remaining auditable. Iterative GPA removes nuisance similarity transformations, the tangent basis expresses 132 independent local shape coordinates, PCA summarizes simulated covariance axes, regularization stabilizes the quadratic metric, and residuals preserve landmark-level differences.

Its strongest design choice is also its main limitation: all reference distributions are synthetic. The system cannot estimate prevalence, group means, uncertainty for a target population, or external validity. It assumes exact landmark correspondence and does not model landmark acquisition error, perspective, expression, occlusion, missingness, or repeated measurements. The MediaPipe-to-68 adapter is approximate, and piecewise-affine image warping can contain seams or artifacts. The analytic shrinkage estimate is fitted to a simulated ensemble and has no external validation. Tangent coordinates are local and may distort large geodesic separations.

For the deterministic pooled Blend example, the conclusion is narrow: after centering, unit scaling, and proper rotation, the synthetic Blend is separated from the pooled synthetic consensus by partial Procrustes 0.06441 and residual RMS 0.00781, while its covariance-weighted tangent distance is 9.759. Its neutral Geometric Displacement Index is 0.46 of 10. The proportional warp visualizes this residual direction without altering the calculation; none of these values evaluates appearance.

Accordingly, the project should be used as an educational and engineering foundation. Any transition to empirical research requires representative de-identified data, measurement reliability, governance, uncertainty propagation, ethical review, and out-of-sample validation. Until those conditions are met, descriptive geometry is the appropriate endpoint.

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